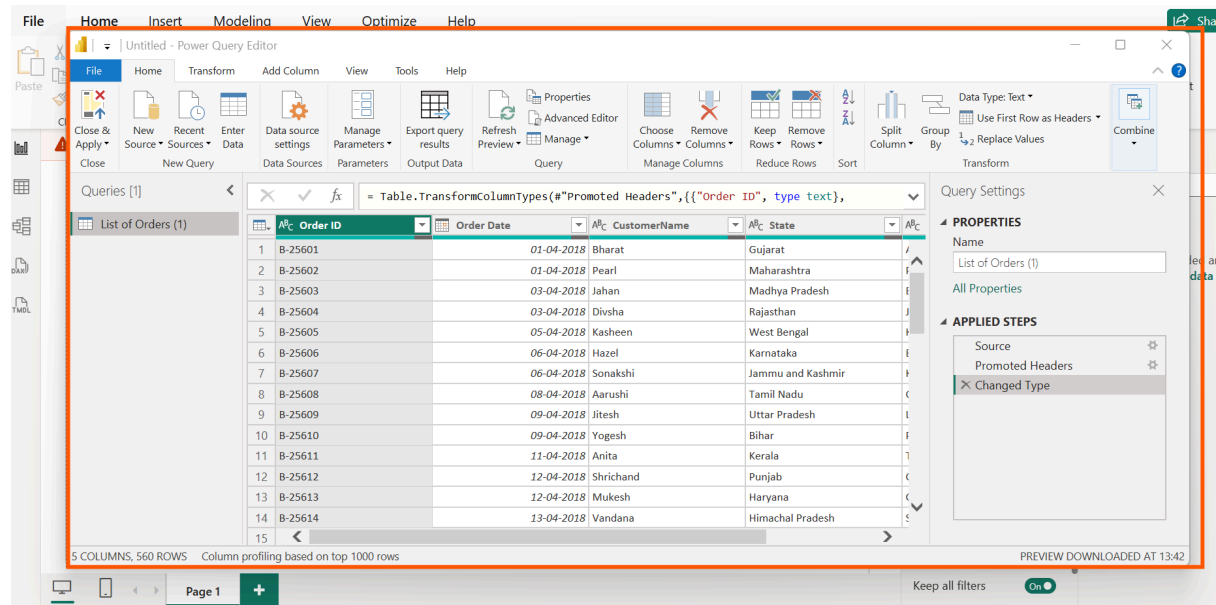
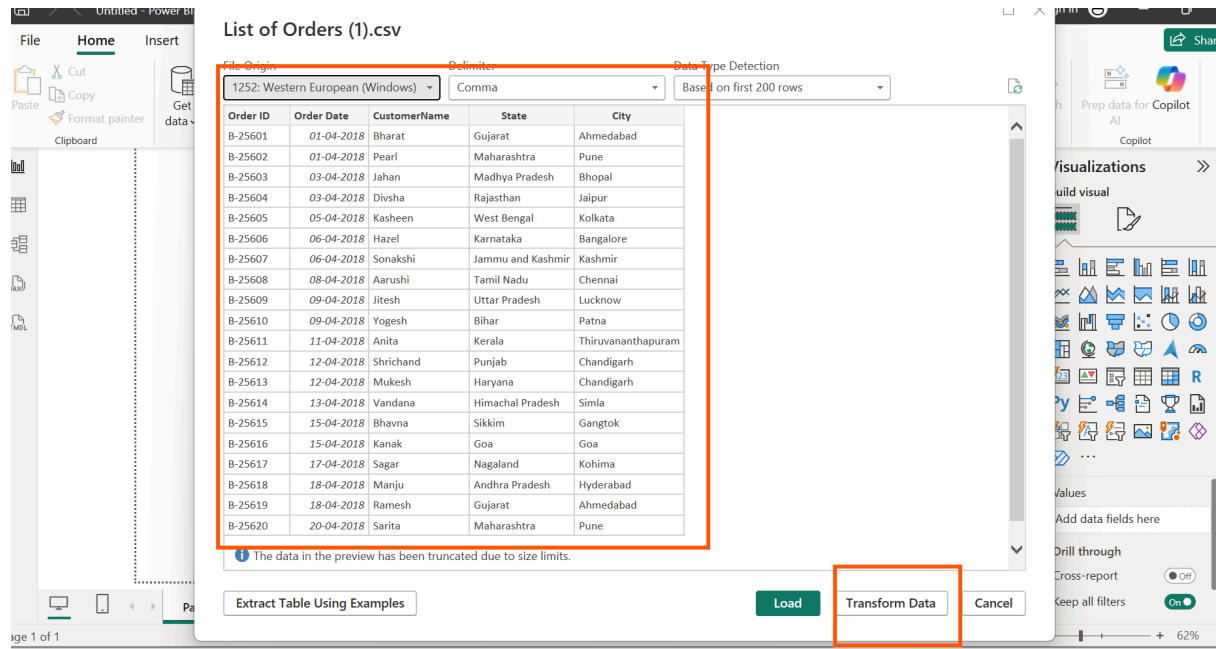
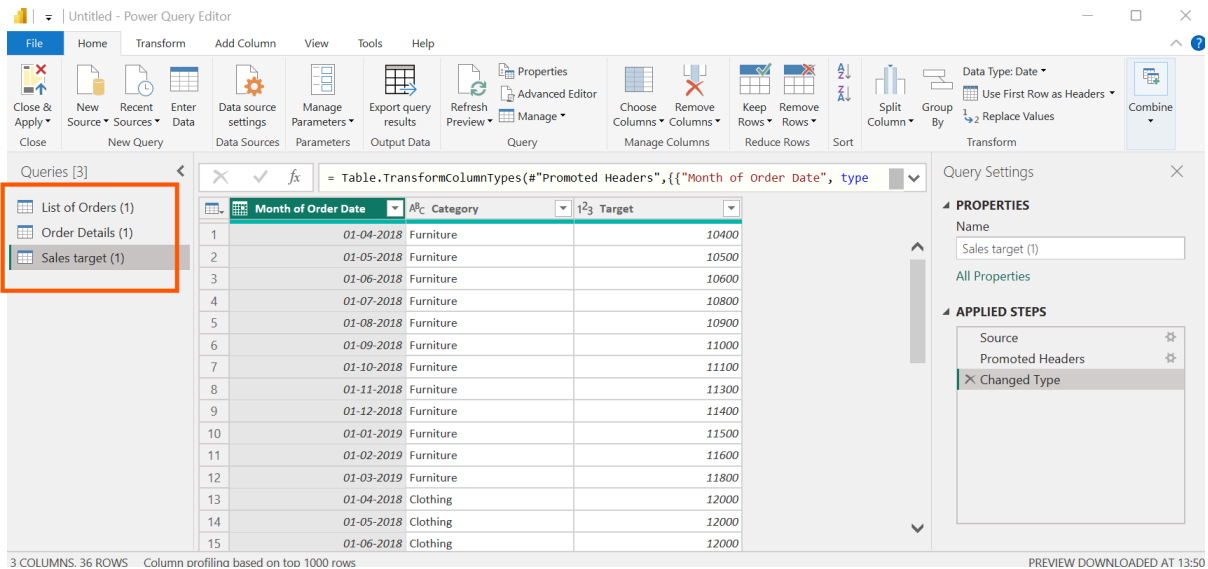


# Data Import & Transformation

## 1. Data Import:

- Import List of Orders.csv into Power BI.
- Open the list of orders in the Power Query Editor using the Transform Data option.

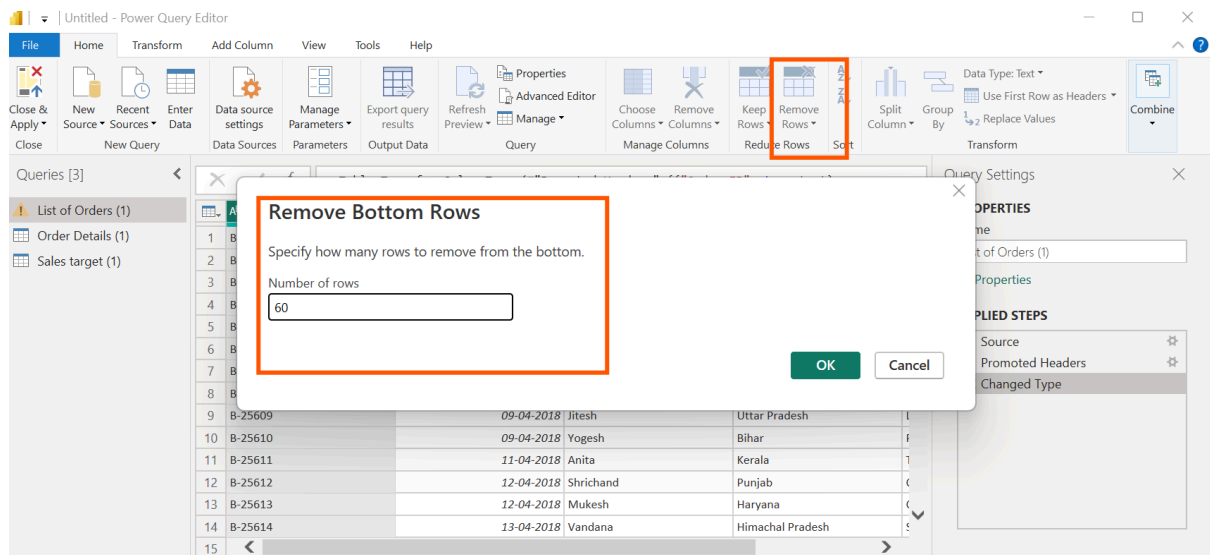




I imported the **List of Orders.csv** into Power BI. Then, I opened the **List of Orders** table in the Power Query Editor using the **Transform Data** option. After that, I imported **Order Details.csv** and **Sales Target.csv** into the Power Query Editor.

## 2. Row Limiting & Data Types:

- Restrict the List of Orders table to the first 500 rows.
- Set the Order Date column to the Date data type.
- Change the Amount and Target columns to fixed decimal numbers.



There were several unwanted rows at the bottom of the **List of Orders** table. Since I only needed the first 500 rows, I removed the extra 60 rows using the **Remove Bottom Rows** option in the Power Query Editor. This restricted the **List of Orders** table to the first 500 rows.

The screenshot shows the Power Query Editor interface. The main data view displays a table with columns: Order ID, Order Date, CustomerName, and State. The 'Order ID' column is highlighted with a red box. In the 'Applied Steps' pane on the right, the step 'Removed Unwanted Bottom Rows' is highlighted with a red box. The 'Query Settings' pane shows the 'Name' as 'List of Orders (1)'.

| Order ID | Order Date | CustomerName | State             |
|----------|------------|--------------|-------------------|
| 487      | 26-03-2019 | Vandana      | Himachal Pradesh  |
| 488      | 26-03-2019 | Bhavna       | Sikkim            |
| 489      | 26-03-2019 | Kanak        | Goa               |
| 490      | 27-03-2019 | Sagar        | Nagaland          |
| 491      | 27-03-2019 | Manju        | Andhra Pradesh    |
| 492      | 27-03-2019 | Ramesh       | Gujarat           |
| 493      | 27-03-2019 | Sarita       | Maharashtra       |
| 494      | 27-03-2019 | Deepak       | Madhya Pradesh    |
| 495      | 28-03-2019 | Monisha      | Rajasthan         |
| 496      | 28-03-2019 | Atharv       | West Bengal       |
| 497      | 28-03-2019 | Vini         | Karnataka         |
| 498      | 29-03-2019 | Pinky        | Jammu and Kashmir |
| 499      | 30-03-2019 | Bhishm       | Maharashtra       |
| 500      | 31-03-2019 | Hitika       | Madhya Pradesh    |

I then renamed the applied step to **"Removed Unwanted Bottom Rows."**

The screenshot shows the Power Query Editor interface. The main data view displays a table with columns: Order ID, Order Date, CustomerName, and State. The 'Order Date' column is highlighted with a red box. In the 'Applied Steps' pane on the right, the step 'Order Date- Date Data Type' is highlighted with a red box. The 'Query Settings' pane shows the 'Name' as 'List of Orders (1)'.

| Order ID | Order Date | CustomerName | State             |
|----------|------------|--------------|-------------------|
| 1        | B-25601    | arar         | Gujarat           |
| 2        | B-25602    | arl          | Maharashtra       |
| 3        | B-25603    | an           | Madhya Pradesh    |
| 4        | B-25604    | sha          | Rajasthan         |
| 5        | B-25605    | heen         | West Bengal       |
| 6        | B-25606    | zel          | Karnataka         |
| 7        | B-25607    | akshi        | Jammu and Kashmir |
| 8        | B-25608    | rushi        | Tamil Nadu        |
| 9        | B-25609    | sh           | Uttar Pradesh     |
| 10       | B-25610    | gesh         | Bihar             |
| 11       | B-25611    | ta           | Kerala            |
| 12       | B-25612    | ichand       | Punjab            |
| 13       | B-25613    | kesh         | Haryana           |
| 14       | B-25614    | Vandana      | Himachal Pradesh  |
| 15       | B-25615    | Bhavna       | Sikkim            |
| 16       | B-25616    | Kanak        | Goa               |
| 17       | B-25617    | Sagar        | Nagaland          |

I set the **Order Date** column to the **Date** data type. Then, I renamed the applied step to **"Order Date – Date Data Type."**

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Whole Number Use First Row as Headers Replace Values Combine

Queries [3] List of Orders (1) Order Details (1) Sales target (1)

fx = Table.TransformColumnTypes(#"Promoted Headers",{{"Order ID", type text},

| Order ID | Amount             | Profit | Quantity |
|----------|--------------------|--------|----------|
| B-25601  | 1.2                | -1148  | 7        |
| B-25601  | \$                 | -12    | 5        |
| B-25601  | 1.2                | -2     | 3        |
| B-25601  | %                  | -56    | 4        |
| B-25602  | Date/Time          | -111   | 2        |
| B-25602  | Date               | -272   | 5        |
| B-25602  | Time               | 1151   | 4        |
| B-25602  | Date/Time/Timezone | 212    | 3        |
| B-25602  | Duration           | -5     | 8        |
| B-25603  | Text               | -60    | 5        |
| B-25603  | True/False         | -30    | 1        |
| B-25603  | Binary             | -166   | 3        |
| B-25603  | Using Locale...    | 5      | 3        |
| B-25603  |                    | 116    | 4        |
| B-25603  |                    | 107    | 6        |
| B-25603  |                    | 12     | 2        |
| B-25603  |                    | 38     | 18       |
| B-25604  |                    | 65     | 17       |

Query Settings

PROPERTIES

Name

Order Details (1)

APPLIED STEPS

Source

Promoted Headers

Changed Type

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Fixed decimal number Use First Row as Headers Replace Values Combine

Queries [3] List of Orders (1) Order Details (1) Sales target (1)

fx = Table.TransformColumnTypes(#"Promoted Headers",{{"Order ID", type text},

| Order ID | Amount   | Profit | Quantity |
|----------|----------|--------|----------|
| B-25601  | 1,275.00 | -1148  | 7        |
| B-25601  | 66.00    | -12    | 5        |
| B-25601  | 8.00     | -2     | 3        |
| B-25601  | 80.00    | -56    | 4        |
| B-25602  | 168.00   | -111   | 2        |
| B-25602  | 424.00   | -272   | 5        |
| B-25602  | 2,617.00 | 1151   | 4        |
| B-25602  | 561.00   | 212    | 3        |
| B-25602  | 119.00   | -5     | 8        |
| B-25603  | 1,355.00 | -60    | 5        |
| B-25603  | 24.00    | -30    | 1        |
| B-25603  | 193.00   | -166   | 3        |
| B-25603  | 180.00   | 5      | 3        |
| B-25603  | 116.00   | 16     | 4        |
| B-25603  | 107.00   | 36     | 6        |
| B-25603  | 12.00    | 1      | 2        |
| B-25603  | 38.00    | 18     | 1        |
| B-25604  | 65.00    | 17     | 2        |

Query Settings

PROPERTIES

Name

Order Details (1)

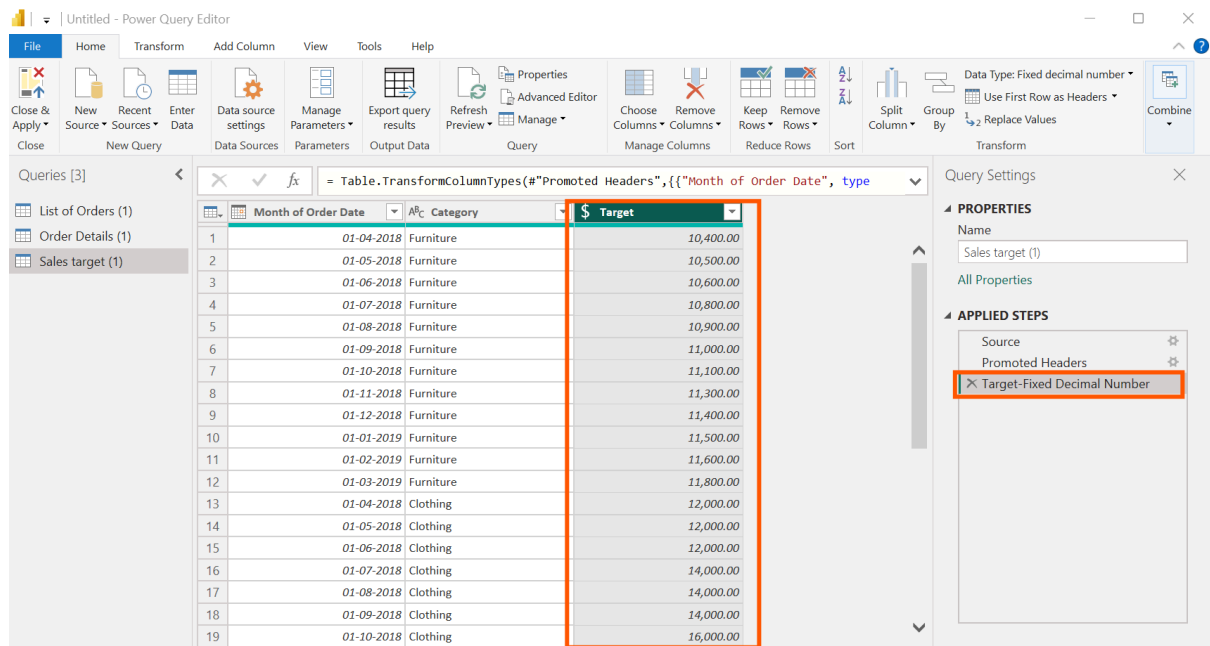
APPLIED STEPS

Source

Promoted Headers

Fixed Decimal Numbers

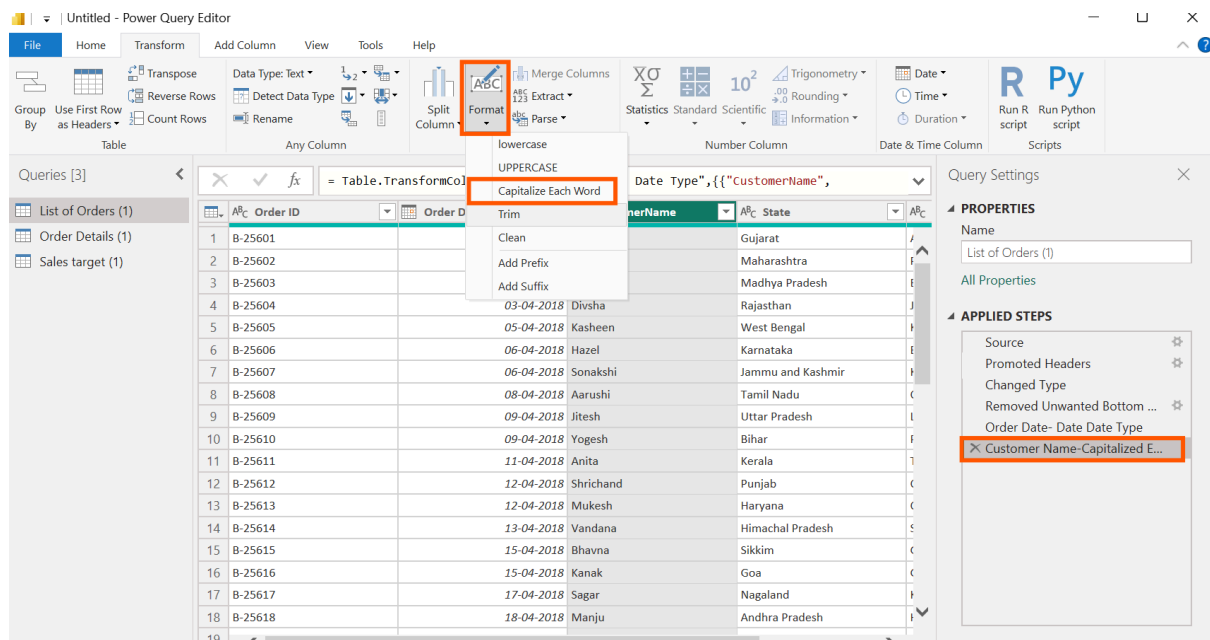
I changed the **Amount** column to the **Fixed Decimal Number** data type.



I changed the **Target** column to the **Fixed Decimal Number** data type.

### 3. Text Formatting:

- Format the CustomerName column to proper case to ensure consistent capitalization.

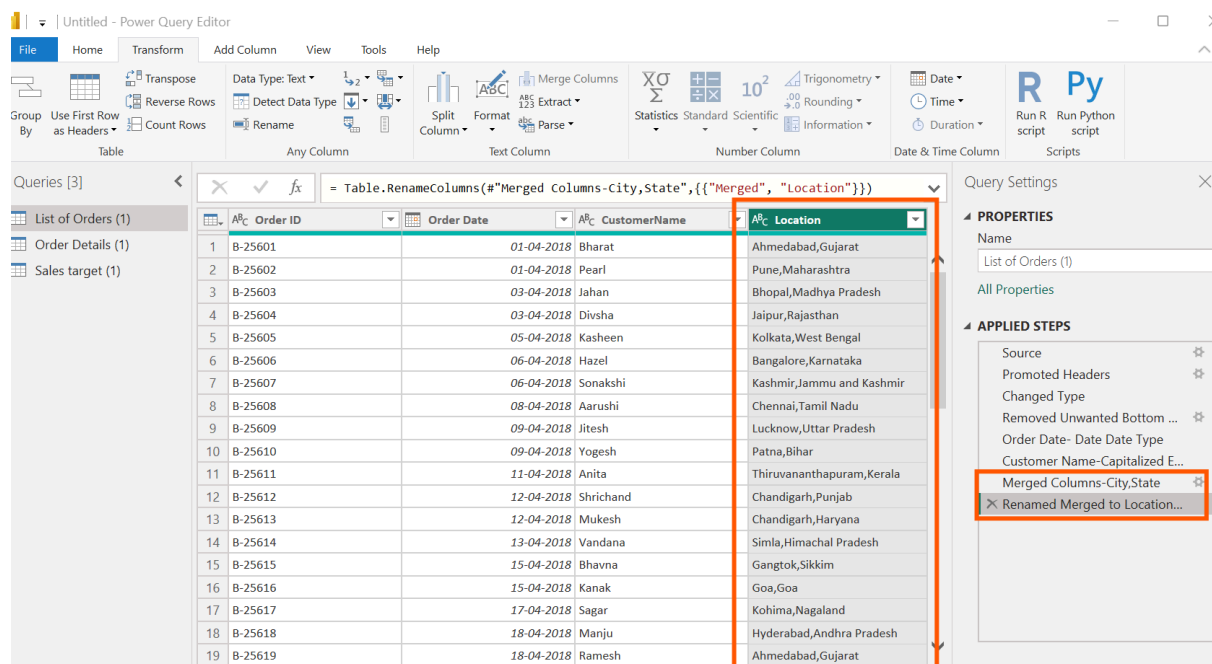
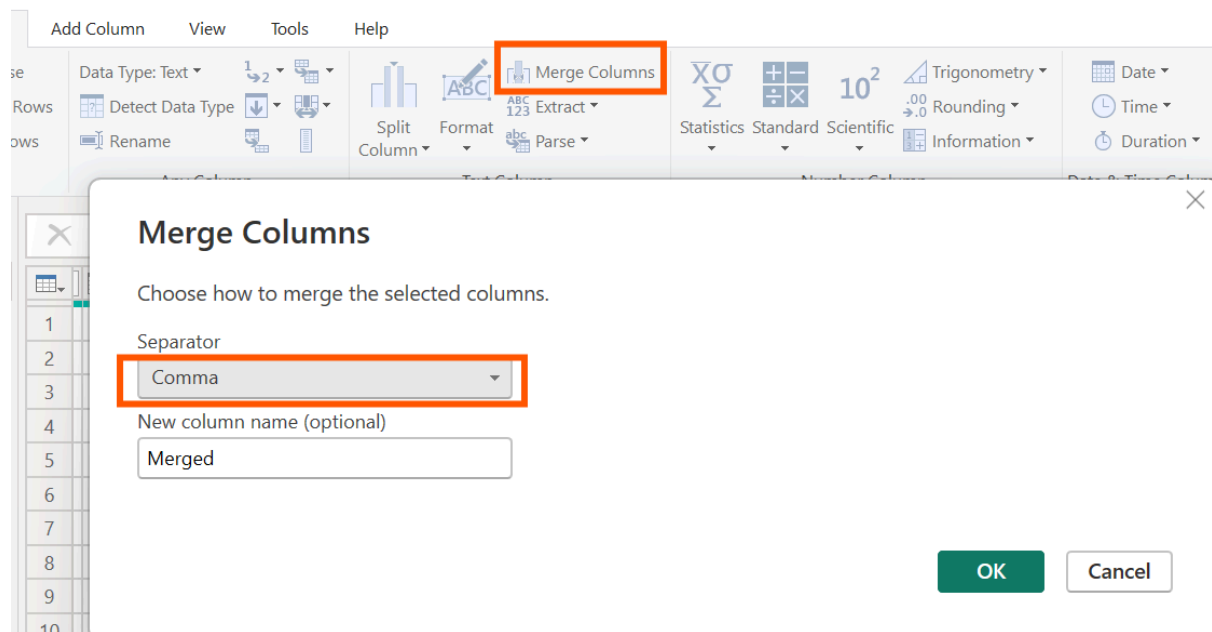


I formatted the **CustomerName** column to **Proper Case** to ensure consistent capitalization.

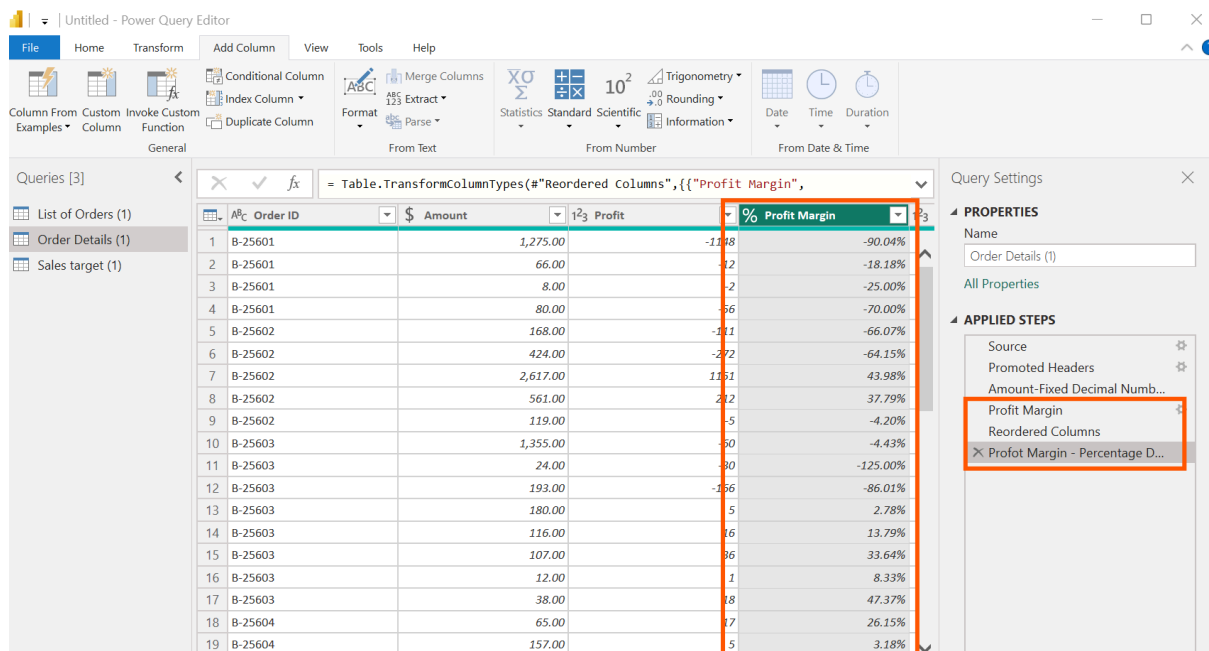
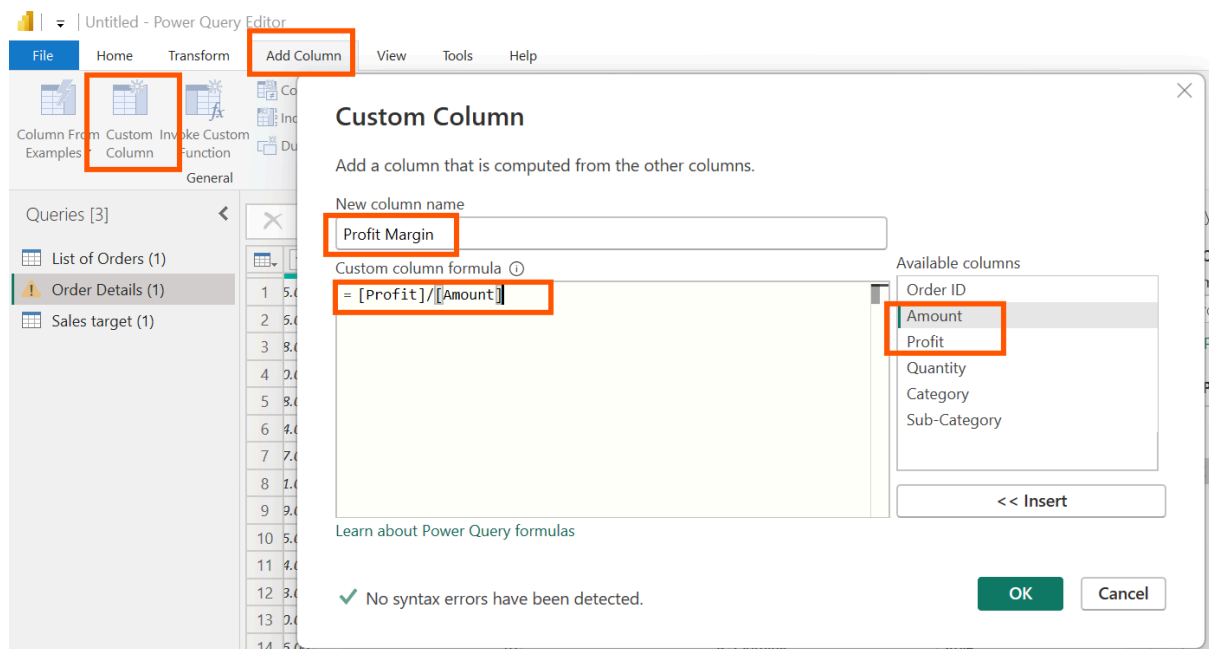
### 4. Column Creation:

- Merge the City and State columns to create a new column named Location in the format: City, State
- Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)

y Editor



I merged the **City** and **State** columns to create a new column named **Location** in the format "**City, State**."



I created a custom column named **Profit Margin** by calculating **Profit / Amount** and formatted the result as a **Percentage**.

## 5. Conditional Column:

- Add a conditional column named Profit Status based on:
  - Profit < 0 → Loss
  - Profit = 0 → Break-Even
  - Profit > 0 → Profit

The screenshot shows the Power Query Editor interface. The top part displays the 'Add Conditional Column' dialog box. The 'New column name' is 'Profit Status'. The 'If' clause is 'Profit' is greater than 0, resulting in 'Profit'. The 'Else If' clause is 'Profit' equals 0, resulting in 'Break-Even'. The 'Else' clause is 'Profit' is less than 0, resulting in 'Loss'. The bottom part shows the resulting data table with columns: Profit, Profit Margin, Profit Status, and Quantity. The 'Profit Status' column is highlighted in orange, showing values 'Loss', 'Break-Even', and 'Profit' based on the 'Profit' values. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, with 'Profit Status- Conditional Col...' and 'Reordered Columns1' highlighted.

**Add Conditional Column**

Add a conditional column that is computed from the other columns or values.

New column name: Profit Status

If: Profit is greater than 0 Then: Profit

Else If: Profit equals 0 Then: Break-Even

Else: Profit is less than 0 Then: Loss

**Table Data:**

|    | Profit | Profit Margin | Profit Status | Quantity |
|----|--------|---------------|---------------|----------|
| 1  | -1148  | -90.04%       | Loss          | 7        |
| 2  | -12    | -18.18%       | Loss          | 5        |
| 3  | -2     | -25.00%       | Loss          | 3        |
| 4  | -56    | -70.00%       | Loss          | 4        |
| 5  | -111   | -66.07%       | Loss          | 2        |
| 6  | -272   | -64.15%       | Loss          | 5        |
| 7  | 1151   | 43.98%        | Profit        | 4        |
| 8  | 212    | 37.79%        | Profit        | 3        |
| 9  | -5     | -4.20%        | Loss          | 8        |
| 10 | -60    | -4.43%        | Loss          | 5        |
| 11 | -30    | -125.00%      | Loss          | 1        |
| 12 | -166   | -86.01%       | Loss          | 3        |
| 13 | 5      | 2.78%         | Profit        | 3        |
| 14 | 16     | 13.79%        | Profit        | 4        |
| 15 | 36     | 33.64%        | Profit        | 6        |
| 16 | 1      | 8.33%         | Profit        | 2        |
| 17 | 18     | 47.37%        | Profit        | 1        |
| 18 | 17     | 26.15%        | Profit        | 2        |

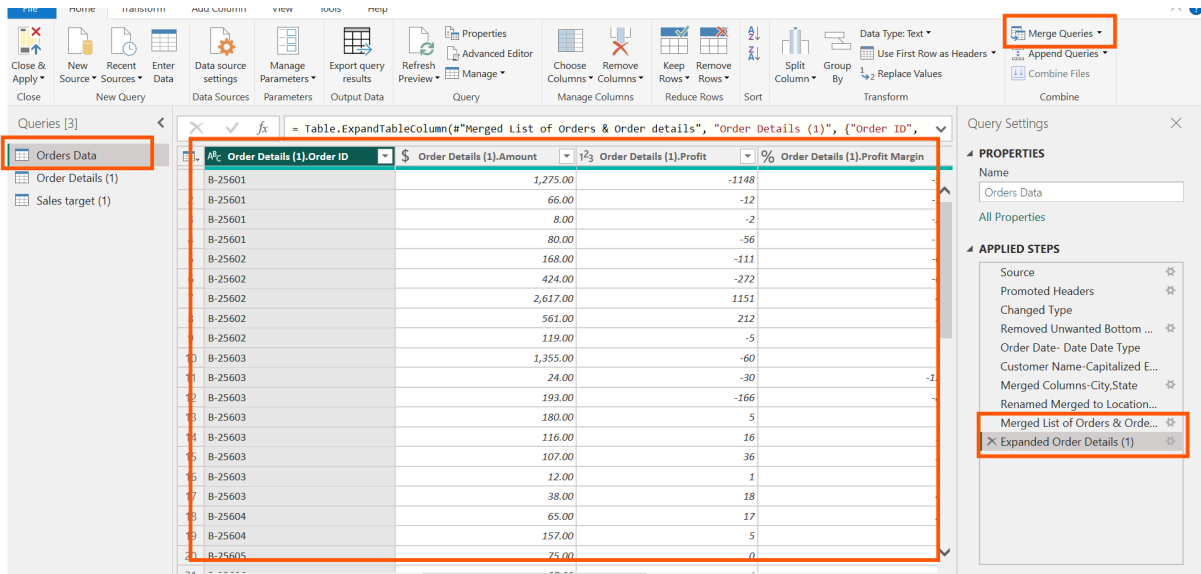
I added a conditional column named **Profit Status** based on the **Profit** values:

- Profit < 0 → Loss
- Profit = 0 → Break-Even
- Profit > 0 → Profit

## 6. Merging Data (Joins)

- Merge the List of Orders and Order Details tables using Order ID.
- Name the merged output table as Orders Data.





I selected the **List of Orders** table in the Power Query Editor, clicked **Home** → **MergeQueries**. then selected **Order Details** as the second table to merge and selected the **Order ID** column in both the **List of Orders** and **Order Details** tables as the matching key. I chose the **Left Outer (all from first, matching from second)** join type and clicked **OK**. I expanded the merged **Order Details** column and selected the required columns to include in the merged table. Finally, I renamed the merged output table to **Orders Data**.

## 7. Handling Missing Data & Duplicate Data

- Identify missing values across all tables.
- Define and apply appropriate strategies such as:
  - Replacing missing values
  - Removing records
  - Retaining nulls with justification
- Identify duplicate rows and define a suitable handling strategy based on business logic.

There were no missing values or duplicate records found in the data.

## 8. Sorting and Filtering Data

- Sort records in the Orders Data table by Order Date in descending order.
- Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).

Query Settings

Name: Orders Data

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Removed Unwanted Bottom ...
- Order Date - Date Type
- Customer Name-Capitalized E...
- Merged Columns-City,State
- Renamed Merged to Location...
- Merged Queries
- Expanded Order Details (1)
- Changed Type1
- Order Date - Descending Order**

| Order ID | Order Date | CustomerName | Location                  | Order Details (1) Order ID |
|----------|------------|--------------|---------------------------|----------------------------|
| 1        | 31-03-2019 | Hetika       | Indore,Madhya Pradesh     | B-26100                    |
| 2        | 31-03-2019 | Hetika       | Indore,Madhya Pradesh     | B-26100                    |
| 3        | 31-03-2019 | Hetika       | Indore,Madhya Pradesh     | B-26100                    |
| 4        | 30-03-2019 | Baishm       | Mumbai,Maharashtra        | B-26099                    |
| 5        | 30-03-2019 | Baishm       | Mumbai,Maharashtra        | B-26099                    |
| 6        | 30-03-2019 | Baishm       | Mumbai,Maharashtra        | B-26099                    |
| 7        | 30-03-2019 | Baishm       | Mumbai,Maharashtra        | B-26099                    |
| 8        | 29-03-2019 | Prnky        | Kashmir,Jammu and Kashmir | B-26098                    |
| 9        | 29-03-2019 | Prnky        | Kashmir,Jammu and Kashmir | B-26098                    |
| 10       | 29-03-2019 | Prnky        | Kashmir,Jammu and Kashmir | B-26098                    |
| 11       | 29-03-2019 | Prnky        | Kashmir,Jammu and Kashmir | B-26098                    |
| 12       | 29-03-2019 | Prnky        | Kashmir,Jammu and Kashmir | B-26098                    |
| 13       | 29-03-2019 | Prnky        | Kashmir,Jammu and Kashmir | B-26098                    |
| 14       | 28-03-2019 | Aharv        | Kolkata,West Bengal       | B-26096                    |
| 15       | 28-03-2019 | Vni          | Bangalore,Karnataka       | B-26097                    |
| 16       | 28-03-2019 | Vni          | Bangalore,Karnataka       | B-26097                    |
| 17       | 28-03-2019 | Aharv        | Kolkata,West Bengal       | B-26096                    |
| 18       | 28-03-2019 | Vni          | Bangalore,Karnataka       | B-26097                    |
| 19       | 28-03-2019 | Aharv        | Kolkata,West Bengal       | B-26096                    |
| 20       | 28-03-2019 | Aharv        | Kolkata,West Bengal       | B-26096                    |

I sorted the records in the **Orders Data** table by the **Order Date** column in **descending order**. This arranged the records from the most recent order date to the oldest order date, making it easier to analyze the latest sales data first.

Query Settings

Name: Orders Data

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Removed Unwanted Bottom ...
- Order Date - Date Type
- Customer Name-Capitalized E...
- Merged Columns-City,State
- Renamed Merged to Location...
- Merged Queries
- Expanded Order Details (1)
- Changed Type1
- Order Date - Descending Order
- Filtered Location-Chennai,Ta...**

| Order ID | Order Date | CustomerName | Location | Order Details (1) Order ID |
|----------|------------|--------------|----------|----------------------------|
| 1        | 22-03-20   | B-26081      | B-26081  | B-26081                    |
| 2        | 22-03-20   | B-26081      | B-26081  | B-26081                    |
| 3        | 22-03-20   | B-26081      | B-26081  | B-26081                    |
| 4        | 22-03-20   | B-26081      | B-26081  | B-26081                    |
| 5        | 22-03-20   | B-26081      | B-26081  | B-26081                    |
| 6        | 22-03-20   | B-26081      | B-26081  | B-26081                    |
| 7        | 14-02-20   | B-26018      | B-26018  | B-26018                    |
| 8        | 14-02-20   | B-26018      | B-26018  | B-26018                    |
| 9        | 09-02-20   | B-26008      | B-26008  | B-26008                    |
| 10       | 09-02-20   | B-26008      | B-26008  | B-26008                    |
| 11       | 09-02-20   | B-26008      | B-26008  | B-26008                    |
| 12       | 09-02-20   | B-26008      | B-26008  | B-26008                    |
| 13       | 15-11-20   | B-25860      | B-25860  | B-25860                    |
| 14       | 21-09-20   | B-25788      | B-25788  | B-25788                    |
| 15       | 11-07-20   | B-25716      | B-25716  | B-25716                    |
| 16       | 23-06-20   | B-25698      | B-25698  | B-25698                    |
| 17       | 23-06-20   | B-25698      | B-25698  | B-25698                    |
| 18       | 23-06-20   | B-25698      | B-25698  | B-25698                    |
| 19       | 23-06-20   | B-25698      | B-25698  | B-25698                    |
| 20       | 23-06-20   | B-25698      | B-25698  | B-25698                    |
| 21       | 23-06-20   | B-25698      | B-25698  | B-25698                    |

I applied a filter to the **Location** column in the **Orders Data** table to display orders from a specific state, such as **Tamil Nadu**. This allowed me to analyze only the orders placed in the selected state and exclude records from other locations.

## 9. Grouping and Aggregating Data

- Duplicate the Order Details table and perform the following:
  - Count of Order ID
  - Average Profit by Category
  - Total Amount by Sub-Category

- Duplicate the Sales Target table and aggregate the total Target by Month of order date.

The screenshot shows the Power BI Desktop interface. The 'Group By' dialog box is open, allowing the user to specify the column to group by and the desired output. The 'Order ID' column is selected for grouping. The 'New column name' is set to 'Count' and the 'Operation' is 'Count Rows'. The 'Basic' tab is selected. The background table shows data for various Order IDs with corresponding amounts and profit percentages.

The screenshot shows the result of the 'Group By' operation. The table now has two columns: 'Order ID' and 'Count'. The 'Count' column shows the total count for each Order ID. The 'Order Details - Duplicate' table is highlighted in the Queries list. The 'Applied Steps' pane on the right shows the 'Count of Order ID' step.

Here, I created a duplicate table from the **Order Details** table. Then, I used the **Group By** option to find the **Count of Order ID**. The total count of Order IDs was **500**. However, since each Order ID contains multiple order details, the Group By operation displayed the detailed

count of records for each individual Order ID.

The screenshot shows the Power BI Desktop interface. The 'Queries' pane on the left lists 'Orders Data', 'Order Details (1)', 'Sales target (1)', and 'Order Details - Duplicate'. The 'Order Details - Duplicate' query is selected. The main view displays a table with two columns: 'Sub-Category' and 'Count'. The table contains 17 rows of data, including 'Bookcases', 'Stole', 'Hankerchief', 'Electronic Games', 'Phones', 'Saree', 'Trousers', 'Chairs', 'Kurti', 'T-shirt', 'Shirt', 'Leggings', 'Tables', 'Printers', 'Accessories', 'Furnishings', and 'Skirt'. The 'Count' column shows values ranging from 1946 to 56861. The 'Group By' button in the 'Transform' ribbon is highlighted with a red box. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' section with 'Tot Amount by Sub-Categories' selected.

| Sub-Category       | Count |
|--------------------|-------|
| 1 Bookcases        | 56861 |
| 2 Stole            | 18546 |
| 3 Hankerchief      | 14608 |
| 4 Electronic Games | 39168 |
| 5 Phones           | 46119 |
| 6 Saree            | 53511 |
| 7 Trousers         | 30039 |
| 8 Chairs           | 34222 |
| 9 Kurti            | 3361  |
| 10 T-shirt         | 7382  |
| 11 Shirt           | 7555  |
| 12 Leggings        | 2106  |
| 13 Tables          | 22614 |
| 14 Printers        | 58252 |
| 15 Accessories     | 21728 |
| 16 Furnishings     | 13484 |
| 17 Skirt           | 1946  |

Here, I used the **Group By** option to find the **Total Amount by Sub-Category**. I grouped the records based on the **Sub-Category** column. Then, I selected the **Amount** column and summarized it using the **Sum** operation. Finally, I obtained the total amount for each sub-category.

The screenshot shows the 'Group By' dialog box in Power BI Desktop. The dialog box is titled 'Group By' and contains the following fields: 'Specify the column to group by and the desired output.', 'Basic' (selected) and 'Advanced' radio buttons, 'Month of Order Date' in the 'Column' dropdown, 'Count' in the 'New column name' field, 'Sum' in the 'Operation' dropdown, and 'Target' in the 'Column' dropdown. The 'OK' button is highlighted with a red box. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' section with 'Tot Target by Month of Order ...' selected.

The screenshot shows the Power BI Desktop interface. The 'Queries' pane on the left lists 'Orders Data', 'Order Details (1)', 'Sales target (1)', 'Order Details - Duplicate', and 'Sales target - Duplicate'. The 'Sales target - Duplicate' query is selected. The main view displays a table with two columns: 'Month of Order Date' and 'Count'. The table contains 12 rows of data, including '01-04-2018', '01-05-2018', '01-06-2018', '01-07-2018', '01-08-2018', '01-09-2018', '01-10-2018', '01-11-2018', '01-12-2018', '01-01-2019', '01-02-2019', and '01-03-2019'. The 'Count' column shows values ranging from 31400 to 43800. The 'Group By' button in the 'Transform' ribbon is highlighted with a red box. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' section with 'Tot Target by Month of Order ...' selected.

| Month of Order Date | Count |
|---------------------|-------|
| 1 01-04-2018        | 31400 |
| 2 01-05-2018        | 31500 |
| 3 01-06-2018        | 31600 |
| 4 01-07-2018        | 33800 |
| 5 01-08-2018        | 33900 |
| 6 01-09-2018        | 34000 |
| 7 01-10-2018        | 36100 |
| 8 01-11-2018        | 36300 |
| 9 01-12-2018        | 36400 |
| 10 01-01-2019       | 43500 |
| 11 01-02-2019       | 43600 |
| 12 01-03-2019       | 43800 |

Here, I used the **Group By** option to find the **Total Target by Month of Order Date**. I grouped the records based on the **Month of Order Date** column. Then, I selected the

**Target** column and summarized it using the **Sum** operation. Finally, I obtained the total target value for each month of the **Order Date**.

## 10. Data Modeling:

- Establish a relationship between:
    - List of Orders and Order Details using Order ID
  - Establish a relationship between:
    - Order Details and Sales Target using Category
2. Use Manage Relationships to ensure relationships are active and correctly defined.

The following tables are shown in the screenshots:

**Orders Data**

| CustomerName | Location         | Order Date    | Order Details... | Order Details ... | Order Details ... | Order Det... |
|--------------|------------------|---------------|------------------|-------------------|-------------------|--------------|
| Bhishm       | Mumbai,Mah...    | 30 March 2019 | Clothing         | B-26099           | 37                | 0.17874...   |
| Vini         | Bangalore,Kar... | 28 March 2019 | Clothing         | -26097            | 5                 | 0.35714...   |
| Vini         | Bangalore,Kar... | 28 March 2019 | Clothing         | -26097            | 8                 | 0.42105...   |

**Order Details (1)**

| Category | Order ID | Profit | Profit Margin   | Profit Status | Quantity | Sub-Categ  |
|----------|----------|--------|-----------------|---------------|----------|------------|
| Clothing | B-25603  | 1      | 0.0833333333... | Profit        | 2        | Hankerc... |
| Clothing | B-25608  | 23     | 0.0894941634... | Profit        | 5        | Hankerc... |
| Clothing | B-25615  | 20     | 0.2941176470... | Profit        | 5        | Hankerc... |

**Sales target (1)**

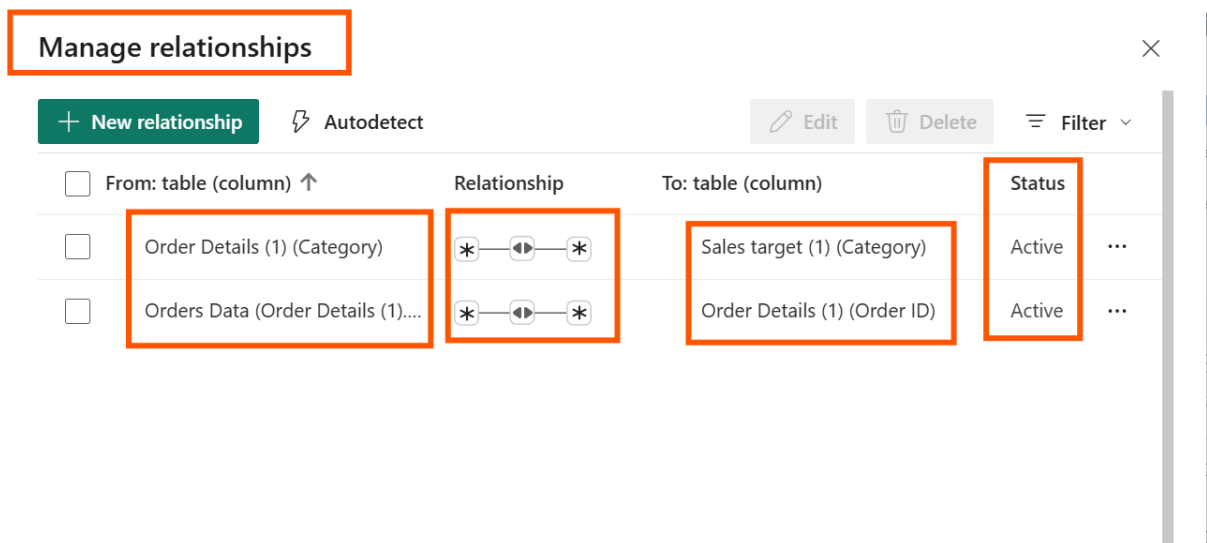
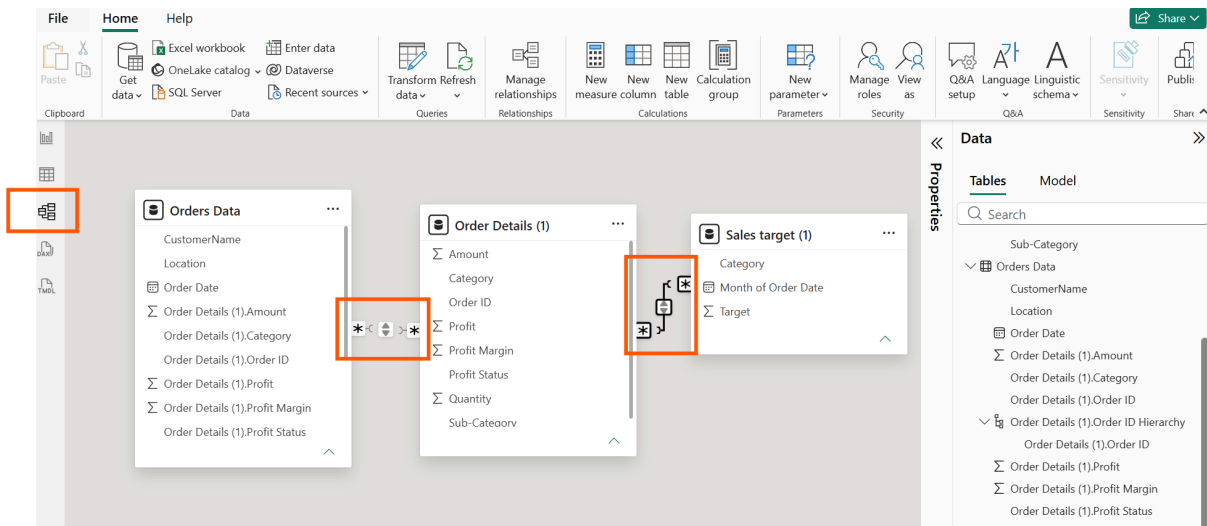
| Category  | Month of Ord... |
|-----------|-----------------|
| Furniture | 01 April 2018   |
| Furniture | 01 May 2018     |
| Furniture | 01 June 2018    |

**Relationship 1: Orders Data to Order Details (1)**

- From table:** Orders Data
- To table:** Order Details (1)
- Cardinality:** Many to many (\*:\*)
- Cross-filter direction:** Both
- ☒ Make this relationship active
- ☐ Assume referential integrity

**Relationship 2: Order Details (1) to Sales target (1)**

- From table:** Order Details (1)
- To table:** Sales target (1)
- Cardinality:** Many to many (\*:\*)
- Cross-filter direction:** Both
- ☒ Make this relationship active
- ☐ Assume referential integrity



Finally, I cleaned all the respective tables using **Power Query** and loaded them into **Power BI**. After loading the data, I used **Data Modeling** to establish relationships between the tables. I created a relationship between the **Orders Data** and **Order Details** tables using the **Order ID** column. I also established a relationship between the **Order Details** and **Sales Target** tables using the **Category** column. Then, I used the **Manage Relationships** option to ensure that all relationships were active and correctly defined.